observIT

Installation Guide

Introduction

ObservIT is provided as a virtual appliance (referred to as vApp) image, which can be deployed on Hyper-V or VMware. If a different hypervisor is required, please contact our support team to request a compatible format.

The vApp image can be downloaded from the following link: [Download ObservIT vApp]

System Requirements

The virtual appliance requires the following system specifications:

Component	Requirement
vCPU	4
Memory (GB)	8
Disk (GB)	45 (30 + 15)

Although internet access is not mandatory, it is recommended to enable HTTP and HTTPS protocols for updates. All collected data will remain stored on-premises, ensuring no data is uploaded externally.

Licensing

ObservIT leverages several open-source tools. Below is a summary of the components and their respective licenses:

Components	Licensing
Ubuntu	https://ubuntu.com/legal/intellectual-property-policy
make	https://www.gnu.org/licenses/gpl-3.0.html
git	https://git-scm.com/about/free-and-open-source
docker engine	https://github.com/moby/moby/blob/master/LICENSE
Python	https://opensource.org/license/python-2-0
Grafana	https://grafana.com/licensing/
Influxdb	https://github.com/influxdata/influxdb/blob/master/LICENSE

Initial Setup

Once the appliance is installed, login with user observit and execute the following command to initialize the vApp setup:

```
setup
```

A menu with the following options will show, choose wisely according to your needs :

Update OS

Updates and upgrades the OS :

1. Select **Update OS** from the menu.
2. Press any key when prompted to upgrade the system.
3. Press any key again to upgrade to a new release if available.

Change Network Settings

Changes network settings via a user-friendly dialog:

1. Select **Change Network Settings** from the menu.
2. Fill in the IP/CIDR, Gateway, and DNS fields.
3. Review the summary and confirm to apply the settings.

Manage Containers

Manages Docker containers (start, stop, restart):

1. Select **Manage Containers** from the menu.
2. Choose a container to manage from the list.
3. Select an action (Start, Stop, Restart) for the selected container.

Containers Housekeeping

Performs housekeeping tasks for Docker containers:

1. Select **Containers Housekeeping** from the menu.
2. The script will perform Docker system pruning and display the results.

Enable DHCP

Enables DHCP:

1. Select **Enable DHCP** from the menu.
2. The script will enable DHCP by removing the static network configuration.

Change Timezone

Changes the system timezone:

1. Select **Change Timezone** from the menu.
2. Choose a timezone from the list to apply it to the system.

Configuration Pre-Requirements

ETERNUS CS8000 :

Every metric that needs to make ssh, make sure you generate a private key and in the destination you use the public key in the authorized keys, then keep the private keys under:

```
ssh-keygen -t rsa -b 2048
```

Choose the location of the private and public key in /opt/observit/collector/keys with no passphrase.

Ensure the vApp have port 22 open to the VLP which will be used to access all the components of CS8000.

The public key must be added to the user used for the login (we assume observit) so, /home/observit/.ssh/authorized_keys owned by observit and with 600 permissions, and the .ssh directory should be 700 with owner observit.

Please remember that the user in CS8000 must be created using:

```
ecs-add-user --user=observit --role=csobserve
```

Depending on the metrics you choose for CS8000 please ensure you configured correctly the sudo permissions:

Metrics:

fs_io	Reports the "svctm", "r_await", "w_await", "r/s" and "w/s" for each filesystem/dm/rawdevice If file cafs_iostat exists under the dir tests it will be used instead of real data. sudo is required: In /etc/sudoers you must have "observit CSTOR = NOPASSWD: /opt/fsc/CentricStor/bin/rdNsdInfos -a"
drives	Reports the drive occupation for each physical library sudo is required: In /etc/sudoers you must have "observit CSTOR = NOPASSWD: /opt/fsc/bin/plmcmd query *"
pvgprofile	Reports for each PVG the following "Total Medias" "Fault Medias" "Inacessible Medias" "Scratch Medias" "Medias with -10%, -20%, -30%, -40%, -50%, -60%, -70%, -80%, -90% and >90% valid LV's" "Total Cap (GiB)" "Total Used (GiB)" sudo is required: In /etc/sudoers you must have "observit CSTOR = NOPASSWD: /opt/fsc/bin/plmcmd query *"
medias	Reports all medias: "Total Medias" "Total Cap GiB" "Total Val GiB" "Val %" "Total Clean Medias" "Total Ina" "Total Fault" sudo is required: In /etc/sudoers you must have "observit CSTOR = NOPASSWD: /opt/fsc/bin/plmcmd query *"
fs	Reports, for each mount point, the "available", "used", "total" filesystem usage in KB for each system
cpu	Reports the cpu "use", "iowait", "load1m", "load5m", "load15m" usage in each system.
mem	Reports the memory "total", "used", "free", "shared", "buff", "avail" usage in MB for each system
fc	Reports the fc throughput in each HBA, including the targets that are virtual drives
net	Reports the acumulated "rx_bytes", "tx_bytes" for each NIC.
vtldirtycache	Reports the dirty cache for the VTL system

Configuration Manual

Systems

The systems section contains a list of systems to be configured. Each system has the following properties:

name: The name of the system.

resources_types: The type of resource to be monitored (eternus_cs8000).

Config

The config section contains the configuration parameters for each resource_type in a system. It has the following properties:

Parameters

user: The username to be used to authenticate on the host we are retrieving metrics from.

host_keys: The location of the host keys if ssh will be used.

poll: The polling interval in minutes.

bastion: The IP address of the bastion host (if used but commonly is the VLP).

Metrics

A list of metrics to be collected from the system. Each metric has a name property.

Available metrics are: resource_type: eternus_cs8000

- name: cpu
- name: mem
- name: fs
- name: fs_io
- name: drives
- name: pvgprofile
- name: medias
- name: vtldirtycache
- name: net
- name: fc

IPs

A list of IP addresses for the system. Each IP has the following properties:

ip: The IP address of the host to observe.

alias: The alias for the IP address (if specified will be presented in the dashboards and graphs).

ip_host_keys: The location of the host keys for this IP.

ip_bastion: The IP address of the bastion host for this IP.

Example

Here is an example of a system configuration located in /opt/observit/collector/config/config.yaml.

At yellow is the section on a default install you don't need to change.

```
systems:
  - name: LAB
    resources_types: eternus_cs8000
    config:
      parameters:
          user: observit
          host_keys: keys/id_rsa
          poll: 1
      metrics:
          - name: fs_io
          - name: drives
          - name: pvgprofile
          - name: medias
          - name: fs
          - name: cpu
          - name: mem
          - name: fc
          - name: net
      ips:
          - ip: 10.1.1.1
            alias: vlp01
            ip_host_keys: keys/127.0.0.1_rsa
          - ip: 192.168.0.1
            alias: icp01
            ip_host_keys: keys/icp01_rsa
            ip_bastion: 10.1.1.1
global_parameters:
    repository: influxdb
    repository_port: 8086
    repository_protocol: tcp
    repository_api_key: TOBEDEFINEDINSETUP
    loglevel: WARNING
    logfile: logs/observit.log
    auto_fungraph: yes
    grafana_api_key: TOBEDEFINEDINSETUP
    grafana_server: grafana
```

This configuration sets up a system named 'LAB' with the resource type 'eternus_cs8000'. It specifies the user, host keys, polling interval and bastion host. It also specifies the metrics to be collected and the IP addresses for the system. Each IP address has its own alias, host keys, and bastion host in case it's needed.

For further assistance, please reach out to our support team.